

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2021

Programme: BE

Full Marks: 100

Course: Engineering Economics

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why an engineer should follow fundamental principles of engineering economy? Explain. 7
- b) You deposited Rs. 50,000 the beginning of each for 7 years. How much money will be in your account at the end of 10th year when rate of interest is 6% compounded quarterly. Make also cash flow diagram. 8
2. a) Write about the types of cost involved in cost accounting? Clarify the concept of cost for business decision making. 7
- b) Determine conventional and modified B/C ratio for the given project. 8
- The cash flows are as follows:
- | | |
|----------------------------|--------------|
| Initial cost | Rs. 400,000 |
| Life of the project | 10 yrs |
| Salvage value | 100,000 |
| Annual benefits | 75000 |
| Annual disbursements (O&M) | 25000 |
| Interest rate | 10% per year |
- Whether the projects is feasible or not?
3. a) Define payback period method. Explain simple and discounted payback period method with a suitable example. 7
- b) The following data have been estimated for two feasible investment machine X and Machine Y having different useful life. If minimum attractive rate of return is 10% choose the best one project using PW method. Use repeatability assumption. 8
- | | Machine X | Machine Y |
|------------------------|-----------|-----------|
| Initial cost | 20,000 | 30,000 |
| Uniform annual benefit | 10,000 | 10,000 |
| Savlage value | 2000 | 1000 |
| Useful life | 3 | 6 |
4. a) An estimation of a new model generator has the following information. Determine, how long will it operate for break-even point? 8

Purchase cost (Rs.)	3,00,000
Annual maintenance cost (Rs.)	8,000
Annual energy generated at full load	12,000 kw
Value of energy generated	Rs. 3/ kw-hr
Salvage value (Rs.)	60,000
MARR	8% per year

- b) What do you mean by ecological footprint? How ecological limits can be overcome to attain sustainable development? 7
5. a) What is the difference between Income statement and cash flow statement? Describe in detail. 7
- b) List out causes of depreciation. A machine costing of Rs 10,000 is estimated salvage value Rs 500 at the end of 5th year. Find depreciation each years by using 8
- Sinking fund
 - Declining balance
 - SOYD method.
6. a) Explain the ratio analyses in detail of following: 7
- Debt ratio
 - Current ratio
 - Total Asset turnover ratio
 - Inventory turnover ratio
- b) What are the objectives of financial statements? Prepare the balance sheet from the following information for year ended 30th December 2015. 8

Land & Building	50,000	Goodwill	10,000
Plant & Machinery	30,000	Creditors	20,000
Outstanding expenses	500	Loan	15,000
Advance Income	500	Investment	20,000
Debtors	10,000	Reserve	6,000
Cash in hand	5,000	Patent	5,000
Prepaid Insurance	1,000	Furniture	4,000
Inventories	25,000	Capital	1,18,000

7. Write short notes on: (Any two) 2×5
- Personal Tax and corporate tax
 - VAT
 - Mutually exclusive project and contingent project